The following text is adapted from Mark Twain's 1876 novel *The Adventures of Tom Sawyer.* Aunt Polly is having dinner with Tom, her mischievous young nephew.

Aunt Polly asked [Tom] questions that were full of guile, and very deep—for she wanted to trap him into damaging revealments. Like many other simple-hearted souls, it was her pet vanity to believe she was endowed with a talent for dark and mysterious diplomacy, and she loved to contemplate her most transparent devices as marvels of low cunning.

1 Mark for Review

As used in the text, what does the word "devices" most nearly mean?

(A) Anxieties

(B) Methods

(C) Technologies

(D) Illustrations

The Niaux Cave paintings—paintings of animal figures found in what is now France and dating from 11,000–17,000 years ago—are sometimes said to be emotionally powerful despite their age, but in fact the paintings are _____ precisely because of their age. It is the link of shared humanity with the artists across so many centuries that makes the Niaux Cave paintings so meaningful.

2 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) neglected

(B) simple

(C) authoritative

(D) moving

The plaice is an ectothermic (cold-blooded) fish, whereas the Pacific bluefin tuna is a regional endotherm; unlike that of ectotherms, its body temperature remains above the ambient water temperature. The basking shark had been classified as a full ectotherm, but researchers Haley R. Dolton et al. found that its body temperature remains 1.0 to 1.5°C above ambient, which is _____ that classification.

3 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) opportune for

(B) subordinate to

(C) incongruous with

(D) indicative of

Many plants can produce and release jasmonic acid (JA), which repulses attacking insects, but there is a reason plants only do so _____: JA suppresses genes involved in leaf expansion and other developmental processes, so deploying it outside the context of an immediate threat is unlikely to yield a net advantage.

4 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) haphazardly

(B) gratuitously

(C) unremittingly

(D) reactively

In their study of European rollers, lesser grey shrikes, and 92 other passerine birds in Iran, researchers used mitochondrial sequences to validate established species boundaries. When a particular data point was _____, they sought corroboration with the bird's nuclear DNA markers and morphological features to ensure the exception was an accurate finding.

5 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) anomalous

(B) tentative

(C) theoretical

(D) preliminary

Humans aren't the only ones who use tools. Other animals also find tools helpful. Boxer crabs use stinging anemones to defend against attack. And despite sometimes being thought of as simple, many birds make clever use of tools as well. Striated herons have been observed using breadcrumbs and other lures to attract fish to the water's surface, making it easier for the herons to catch them.

6 Mark for Review

Which choice best describes the function of the underlined sentence?

(A) It emphasizes that boxer crabs are smarter than birds.

(B) It provides one example of tool use in a nonhuman animal.

(C) It explains that humans use tools for many different tasks.

(D) It argues that boxer crabs use tools more often than other animals do.

How does the main melody in a song like "Irreplaceable" by Beyoncé (2006) compare to that in an older song like "All Shook Up" by Elvis Presley (1956)? In a 2024 study, Marcus Pearce and his coauthor analyzed the complexity of the main melodies in the five most popular songs each year from 1950 to 2023—including "Irreplaceable" and "All Shook Up"—according to *Billboard*'s year-end singles charts. The researchers considered several features of complexity and concluded that over the decades, main melodies in popular music in the US have become less complex.

7 Mark for Review

Which choice best states the main purpose of the text?

(A) To describe a method used by researchers to evaluate the complexity of melodies

(B) To compare two popular songs that were released many years apart

(C) To provide an overview of the history of *Billboard*'s year-end singles charts

(D) To discuss research that reveals a change over time in US popular music

In *The Jazz Continuum*, LaTasha Barnes, Tyedric Hill, and other dancers seamlessly interweave recent dance styles, including hip-hop and voguing, with elements of the Lindy Hop, a popular swing dance of the 1930s with roots in the Black American community of Harlem, New York. Inspired by Barnes's curiosity about the similarities between the Lindy Hop and these seemingly unrelated dance styles, the show's choreography makes visible how intertwined their histories actually are. The show is therefore a testament to the Lindy Hop's enduring influence on American culture.

8 Mark for Review

Which choice best states the function of the underlined portion in the text as a whole?

(A) It addresses the reasons why the historical connections between the dance styles mentioned earlier in the text have been generally overlooked.

(B) It provides evidence for the text's claim that the Lindy Hop has had a greater cultural impact than the other dances featured in *The Jazz Continuum* have had.

(C) It identifies the motivation for *The Jazz Continuum*'s choreographic approach that is described earlier in the text.

(D) It identifies the specific ways in which the Lindy Hop is similar to the contemporary dance styles discussed in the text.

Companies involved in methane distribution include pipelines among their assets. But methane distribution is a very carbon-intensive industry, so as government agencies increase regulations on carbon emissions, demand for methane falls and the pipelines will eventually, or even suddenly, become liabilities as reduced methane prices make it more difficult to recover the expense of maintaining such equipment.

9 Mark for Review

What claim does the text make about the demand for methane?

(A) Companies involved in methane distribution can benefit from a decline in the demand for methane.

(B) The regulations on carbon emissions are factors in the demand for methane.

(C) The demand for methane follows a predictable annual cycle.

(D) Companies involved in methane distribution are not affected by the demand for methane.

Claudine at School is a 1900 novel by Colette, originally written in French. The narrator is a fifteen-year-old girl living in a small, rural town. She presents herself as having a strong emotional attachment to the surrounding forests: _____

10 Mark for Review

Which quotation from a translation of Claudine at School most effectively illustrates the claim?

(A) "I heard a wolf once, at the beginning of winter, while I was picking up beech-nuts—those nice, oily little beech-nuts that tickle your throat and make you cough."

(B) "And then there are my favourites, the great woods that are sixteen and twenty years old. It makes my heart bleed to see one of those cut down."

(C) "Green meadows make rifts in [the woods] here and there, so do little patches of cultivation. But these do not amount to much, for the magnificent woods devour everything. As a result, this lovely region is atrociously poor and its few scattered farms provide just the requisite number of red roofs to set off the velvety green of the woods."

(D) "If I had a Mamma, I know very well that she would not have let me stay [in Montigny] twenty-four hours. But Papa—he doesn't notice anything and doesn't bother about me."

Defensive Behavior and Reproductive Traits of Select Bird Species

Scientific name	Common name	Broken-wing display?	Incubation (days)	Incubation duty	Max broods per year
Bucephala islandica	Barrow's goldeneye	No	34	1 parent	1
Haematopus bachmani	black oystercatcher	No	33	2 parents	1
Zenaida macroura	mourning dove	Yes	15	2 parents	7
Pooecetes gramineus	vesper sparrow	Yes	13	1 parent	3

In an extensive review of existing literature, Léna de Framond and team cataloged the prevalence of broken-wing display—a defensive behavior observed in *Charadrius semipalmatus* (semipalmated plover) and many other species—throughout the Aves class. Documentation of the display in 285 species across 52 families suggests the behavior likely evolved independently multiple times, prompting the team to consider ecological and life-history characteristics with hypothesized associations to the behavior's emergence, including traits related to reproduction investment and future reproduction potential. Based on their review of those traits, the team concluded that _____

11 Mark for Review

Which choice most effectively uses data from the table to complete the conclusion?

(A) among species with more than one parental incubator, the use of broken-wing display is associated with greater incubation duration.

(B) broken-wing display is most often observed in species with less opportunity to reproduce in a year due to longer incubation periods.

(C) incubation duration and capacity for multiple broods are more strongly associated with the use of broken-wing display than the number of parental incubators is.

(D) capacity for multiple broods, number of parental incubators, and incubation duration are equally associated with the use of broken-wing display.

Hypothesizing that lullabies, characterized by their less steady beat, are universally calming to infants, Constance M. Bainbridge and colleagues played a lullaby sung in the Selk'nam language and a non-lullaby sung in the Chuukese language to a group of infants. The team found that the infants' heart rates and pupil size both decreased more during the lullaby than during the non-lullaby. Since a decrease in heart rate is associated with relaxation, the team concluded that the lullaby relaxed the infants. However, noting that reduced heart rate can also be associated with increased attention, one critic argues that instead, the lullaby simply attracted the infants' attention.

12 Mark for Review

Which finding, if true, would most directly weaken the critic's claim?

(A) Parents of infants in the study preferred the Selk'nam lullaby over the Chuukese non-lullaby

(B) Increased attention to a stimulus is usually accompanied by an increase in pupil size.

(C) Infants in the study had no familiarity with either the Selk'nam language or the melody of the Ona lullaby.

(D) More frequent blinking has also been found to be a reliable indication of attention.

Students in a biology class investigated why individual house mice (*Mus musculus*) can differ from one another in their responsiveness to potential rewards. The students compared wild-type mice and knockout mice, which are mice with specific genes deactivated, when mice of each type were presented with the same reward system for particular behaviors. Finding that knockout mice with the gene *Cartpt* deactivated tended to show less sensitivity to rewards than did wild-type mice, the students concluded that differences in reward responsiveness among house mice in nature are solely attributable to variations in the level of expression of *Cartpt*.

13 Mark for Review

Which finding, if true, would most directly weaken the students' conclusion?

(A) The mice with *Cartpt* deactivated were identical to the wild-type mice except with regard to reward responsiveness.

(B) The level of expression of *Cartpt* does not appear to affect the functioning of any other genes in house mice.

(C) A sampling of house mice captured in natural settings shows that individual mice can differ from one another in the level of expression of *Cartpt*.

(D) Some wild-type mice were very similar to the knockout mice with regard to reward responsiveness but showed a wide variety of levels of expression of *Cartpt*.

Humans are overwhelmingly right-hand dominant (85–90%) and can thus be said to exhibit strong population-level right handedness (PLRH). Among studies of nonhuman primates, G. Haas's 1958 study of captive gorillas purported to show PLRH, while Jane Goodall's 1963 study of wild chimpanzees did not. Overall, the studies claiming PLRH in nonhuman primates find much lower incidences of right-handedness than humans exhibit, but it's worth noting that studies of captive primates tend to show significantly greater incidences of right-handedness than studies of wild primates do, therefore raising the possibility that

_____

14 Mark for Review

Which choice most logically completes the text?

(A) the greater exposure to humans among the captive gorillas induced them to acquire more right-handed behaviors.

(B) the number of individuals in the study of wild chimpanzees is insufficient to preclude the claim that nonhuman primates exhibit PLRH

(C) the apparent discrepancy between the studies' results may be partly attributable to the 1958 study using different criteria when assessing hand dominance than were used in the 1963 study.

(D) the number of individuals in the study of captive gorillas is insufficient for a robust claim regarding evidence of PLRH.

Research such as the 2012 study of mammals by Soledad Lucía Uran and colleagues has shown that noise from human activity, like traffic on a busy highway, has significant effects on animals, and many governments require studies of the potential noise effects on wildlife before approving highway construction projects. Advocates for one such project in an area home to both mammals and birds argue that since previous research has shown that birds are, on average, less affected by such noise than mammals are, the effect of the proposed project on birds does not need to be studied. Conservationists counter that basing such a decision on average effects is problematic because _____

15 Mark for Review

ABC

Which choice most logically completes the text?

(A) the effects of noise on birds have been more extensively documented than the effects on mammals have.

(B) the average for the effect of noise on birds is based on a much larger number of distinct species than theaverage for the effect of noise on mammals is.

(C) addressing the effects of noise on birds is likely to help mitigate some of the effects on mammals.

(D) the severity of the effect of noise on some species of birds could be much higher than the average effect.

The 1913 Armory Show in New York City was a groundbreaking art exhibition planned by a collective of 28 artists, among them Mahonri Young and Maurice B. Prendergast. By this time, many of these artists _____ tired of the staid realism that dominated American art at the time; the Armory Show aimed to jolt the artistic establishment awake with bold, experimental modernism.

16 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) will have grown

(B) grow

(C) are growing

(D) had grown

How a folktale begins depends largely on the language in which it originates. Many folktales in the Algerian Arabic language open with a phrase that roughly translates to "I've told you what's coming." Tales in _____ the other hand, often begin with "once upon a time."

17 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) English: on

(B) English, on

(C) English. On

(D) English—on

A recipient of the Nobel Prize in Chemistry and the Golden Plate Award of the American Academy of Achievement, _____ known as femtochemistry when he devised an optical spectroscopy technique that allowed researchers to study ultrafast chemical reactions occurring within molecules.

18 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) chemist Ahmed Zewail expanded the limits of measurability and introduced a new scientific discipline

(B) the result of chemist Ahmed Zewail's expansion of the limits of measurability was the introduction of a new scientific discipline

(C) the limits of measurability were expanded by chemist Ahmed Zewail who introduced a new scientific discipline

(D) the introduction of a new scientific discipline by chemist Ahmed Zewail led to the expansion of the limits of measurability

In Virginia Woolf's *To the Lighthouse* (1927), the central action around which the novel ostensibly revolves—the Ramsay family's long-promised trip to visit a lighthouse—is deferred for nearly a decade and proves anticlimactic when, in the novel's final section, it finally _____ Woolf deliberately eschews plot-driven development, thus diverging from her Victorian predecessors, reflects her oft-stated conviction that the most significant events in human life occur not in the external world but within consciousness itself.

19 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) happens.

(B) happens. That

(C) happens:

(D) happens that

The 1969 song “Boston Bernie” by Dexter Gordon is a famous jazz composition. Because _____ harmonic structure from an earlier song (Jerome Kern's “All the Things You Are” from 1939) while introducing new melodic elements, Gordon's piece is what's called a “contrafact.”

20 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) they borrow they're

(B) it borrows its

(C) it borrows it's

(D) they borrow their

Drought avoidance mechanisms—morphological adaptations that plants undergo when exposed to drought conditions—take a variety of forms. The Osage orange plant decreases the size of its leaves to reduce water loss through _____ in the soybean plant, drought triggers an elongation of the plant's primary root that increases water uptake.

21 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) transpiration. For instance, while

(B) transpiration, for instance, while

(C) transpiration; for instance,

(D) transpiration, for instance,

The Sydney Desalination Plant in Sydney, Australia, uses reverse osmosis to produce fresh water from salt water. _____ this is not the only way to desalinate (remove salt from) water. The Yanbu Desalination Plant in Yanbu, Saudi Arabia, uses a method called multistage flash distillation.

22 Mark for Review

Which choice completes the text with the most logical transition?

(A) However,

(B) In other words,

(C) As a result,

(D) Likewise,

Within a given ecosystem, energy tends to transfer across what ecologists, such as Veronica Andreo, call trophic levels. _____ primary producers, like plants, at the first trophic level provide energy (i.e., food) for herbivores, like mosquito larvae, at the second level; those at the second level provide energy for carnivores and omnivores, like small birds, at the third; and so on, up the food chain

23 Mark for Review

Which choice completes the text with the most logical transition?

(A) For example,

(B) By contrast,

(C) However,

(D) Likewise,

In 1965, Morrie Turner's comic strip *Wee Pals* became the first serial comic by a Black cartoonist to be syndicated, appearing in papers nationwide. _____ many other strips by Black cartoonists were syndicated, including Stephen Bentley's *Herb and Jamaal* in 1989 and Keith Knight's *The Knight Life* in 2008.

24 Mark for Review

Which choice completes the text with the most logical transition?

(A) Later,

(B) That is,

(C) For example,

(D) Previously,

While researching a topic, a student has taken the following notes:

- Christina Ying is a Filipina American playwright.
- *There's the Moon and Then There's You* is a play by Ying.
- It premiered in 2015 at the Bindlestiff Studio in San Francisco.
- It tells the story of a NASA astronaut named David.
- In the play, David aspires to be selected for a mission to the Moon.

25 Mark for Review

The student wants to provide a brief overview of what happens in There's the Moon and Then There's You. Which choice most effectively uses information from the notes to accomplish this goal?

(A) In *There's the Moon* and *Then There's You,* a NASA astronaut named David aspires to be selected for a mission to the Moon.

(B) *There's the Moon* and *Then There's You* is a 2015 play that premiered at the Bindlestiff Studio in San Francisco.

(C) The 2015 play *There's the Moon* and *Then There's You* tells the story of Christina Ying, a Filipina American playwright.

(D) Ying's play premiered in 2015 at the Bindlestiff Studio, which is located in San Francisco.

While researching a topic, a student has taken the following notes:

- Dinosaur fossil specimens can be found at science museums all over the world.
- A dinosaur fossil specimen nicknamed Sue is housed at the Field Museum of Natural History in Chicago, Illinois.
- *Sue, a Tyrannosaurus,* lived in the Late Cretaceous period.
- A dinosaur fossil specimen nicknamed Doyle is housed at the American Museum of Natural History in New York, New York.
- Doyle, a *Triceratops*, lived in the Late Cretaceous period.

The student wants to emphasize a similarity between the two dinosaur fossil specimens. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) The Field Museum of Natural History is home to Sue, a *Tyrannosaurus* fossil specimen from the Late Cretaceous period.

(B) Both the *Tyrannosaurus* fossil specimen Sue and the *Triceratops* fossil specimen Doyle lived in the Late Cretaceous period.

(C) While the *Tyrannosaurus* fossil specimen Sue is in Chicago, Illinois, the *Triceratops* fossil specimen Doyle is in New York, New York.

(D) Dinosaur fossil specimens can be found at science museums all over the world, including in Chicago, Illinois, and New York, New York.

While researching a topic, a student has taken the following notes:

· The French Republican calendar replaced the Gregorian calendar in France from 1793 to 1805.
· Both calendars are divided into twelve months.
· The timelines for the months don't match. The fourth month of the French Republican calendar, Nivôse, begins during December, the twelfth month of the Gregorian calendar.
· The eleventh month of the French Republican calendar, Thermidor, begins during July, the seventh month of the Gregorian calendar.

27 Mark for Review

The student wants to emphasize a difference between the French Republican and Gregorian calendars. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) The French Republican calendar, which replaced the Gregorian calendar in France from 1793 to 1805, has twelve months, including the months of Nivôse and Thermidor.

(B) Used in France from 1793 to 1805, the French Republican calendar includes the months of Nivôse and Thermidor (the fourth and eleventh months of the year).

(C) The timelines for the months of the two calendars don't match; for example, the fourth month of the French Republican calendar begins during the twelfth month of the Gregorian calendar.

(D) The twelfth and seventh months of the Gregorian calendar are December and July, respectively.

While *higaki* (cypress fence) and other traditional sashiko patterns are visually striking, the repeated stitches in this needlework technique also _____ textiles with resilience. Developed in the Edo period (1603–1868) in Japan to mend clothing and domestic textiles, sashiko uses tightly spaced rows of stitching to beautify and reinforce worn cloth.

1 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) belie

(B) depict

(C) imbue

(D) regale

Hydrogen production via water electrolysis is an emerging industry whose water usage relative to that of other industrial sectors is negligible. The industry's growth is thus unlikely to _____ water scarcity globally, but it could do so in South Korea: competition from other sectors for already scarce water resources in South Korea precludes its achievement of hydrogen self-sufficiency.

2 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) assuage

(B) rescind

(C) extricate

(D) exacerbate

Although it would be overstating the case to characterize the effects of oil shocks—such as the 37% rise in oil prices between November 1947 and January 1948—as merely _____, Gbadebo Oladosu et al. have shown that the effects of oil shocks on gross domestic product (GDP) can be fairly modest. In the United States, for instance, recent oil shocks exerted only a slightly negative effect on GDP.

3 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) detrimental

(B) idiosyncratic

(C) nominal

(D) derivative

Accounts known as popular microhistories tell sweeping, century-spanning stories of ordinary objects like cotton and fish. _____ this approach with that of traditional microhistories, which undertake painstaking historical reconstructions of singular incidents, may seem unlikely, but both are guided by the idea that seemingly inconsequential subjects can reveal larger historical processes in surprising ways.

4 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) Instituting

(B) Reconciling

(C) Contrasting

(D) Supplanting

Positive expressive writing is self-reflective writing aimed at exploring positive topics and feelings like satisfaction. Techniques of positive expressive writing include one called "three good things," which involves writing down three good things that happen each day, and "resource diary," which involves writing about internal strengths and external support systems. In a large-scale analysis, Lauren Hoult and colleagues determined that while different techniques produce varying outcomes, they all tend to have a more consistent effect on well-being and positive outlook than on physical health and stress.

5 Mark for Review

Which choice best describes the overall structure of the text?

(A) It gives an overview of positive expressive writing and then states a research finding about this type of writing.

(B) It identifies several positive emotions and then asserts that positive expressive writing should be done daily to amplify those emotions.

(C) It describes one popular technique of positive expressive writing and then names the techniques researchers have found to be most useful.

(D) It introduces a study about positive expressive writing and then explains that the researchers are planning a follow-up study.

The following text is from William Carlos Williams's 1925 creative nonfiction book *In the American Grain.*

History follows governments and never men. It portrays us in generic patterns, like effigies or the carvings on sarcophagi, which say nothing save, of such and such a man, that he is dead. That's history. It is concerned only with the one thing: to say everything is dead. Then it fixes up the effigy: there that's finished. Not at all. History must stay open; it is all humanity. Are lives to be twisted forcibly about events, the mere accidents of geography and climate?

6 Mark for Review

Which choice best describes the function of the underlined sentence in the text as a whole?

(A) It identifies a flaw in human nature that historians have overlooked.

(B) It emphasizes a characteristic of the discipline of history that Williams condemns.

(C) It highlights a controversy among historians that Williams criticizes.

(D) It raises a question about the practice of history that Williams regards as unresolved.

Text 1

To discern the ecological role that blackberry plants, or *Rubus fruticosus* L. aggregate, serve for species of foraging insects, many of which are in decline, Annabel Squire and team recorded about two thousand insect visits to *R. fruticosus* (between 11 a.m. and 4 p.m.) during the plant's flowering period. They observed visits from a diverse range of species, including *Chloromyia formosa and Apis mellifera.*

Text 2

Researchers' focus on diurnal invertebrate pollinators has resulted in a decided lack of data on moth pollination, but using infrared detection equipment, Max Anderson et al. assessed how diurnal and nocturnal insects separately contribute to *R. fruticosus* pollination. They found that although 83% of all insect visits occurred during the day, pollen deposition rates were much higher during moths' nocturnal foraging.

7 Mark for Review

Based on the texts, how would the author of Text 2 most likely respond to the information about Squire and team's study as presented in Text 1?

(A) By agreeing that the study's findings are helpful with the caveat that the study should have included more species of plants besides *R. fruticosus*

(B) By claiming that the observations by Squire and team might not include an important category of interactions between invertebrate pollinators and *R. fruticosus*

(C) A. By suggesting that Squire and team's conclusions were undermined by a failure to accurately distinguish diurnal pollinator visits from nocturnal pollinator visits in their data

(D) A. By criticizing the study for failing to account for the foraging behavior of vertebrate animals that pollinate *R. fruticosus*

Condemned by colonial officials in Zimbabwe as ruinous to rangelands and forests, goats have often been vilified as agents of environmental degradation. Scrutinizing the putative role of the goat in narratives of ecological decline, researchers Cara Loomis and Carol Kerven suggest that such narratives _____. Because goats are highly resilient, they often inhabit harsh landscapes that can no longer support other livestock species. As a result, they are frequently condemned for damage wrought over extended periods by other factors like overcultivation and soil erosion.

8 Mark for Review

In context, which choice best completes the main argument presented in the text?

(A) may confuse an effect for a cause

(B) tend to generalize from isolated incidents

(C) are largely based on hearsay

(D) may underestimate their subject's insatiability

High-reward flowers that produce relatively high volumes of nectar are thought to be especially attractive to green-fronted lancebills and other hummingbird species. But when María A. Maglianesi and team monitored hummingbird visits to flowering plants, including *Psammisia ramiflora*, at Las Nubes Project and other sites in Costa Rica, they found that visit frequencies weren't influenced by nectar traits. Rather, physical compatibility of flower corollas (collections of petals) and hummingbird bills more strongly predicted hummingbird-plant interaction patterns.

9 Mark for Review

Which statement about hummingbird preferences is most strongly supported by the text?

(A) Hummingbirds consistently prefer plant species whose flowers have physical traits that are compatible with their bills but offer relatively low nectar rewards.

(B) Hummingbirds' preferences are primarily driven by trait-matching between their bills and flower corollas, though nectar qualities are a secondary factor driving their preferences.

(C) A floral trait found to influence which plant species hummingbirds prefer may result in certain hummingbird species visiting flowers with relatively low nectar rewards over those with higher nectar rewards.

(D) Physical compatibility between flower corollas and hummingbird bills becomes an important factor that influences hummingbird preferences when hummingbirds are uncertain about which plants produce high-reward nectar.

Orientation of Paired Leaves in Grapevines and Related Species

Species	Total leaf pairs examined	Pairs with opposite-side	Pairs with same-side	Ratio (n to 1)
Vitis vulpina	155	108	47	2.30
Vitis labrusca	383	287	96	2.99
Vitis coignetiae	58	37	21	1.76
Vitis riparia	843	646	197	3.28

Many plants have leaves that are larger on one side of their long central axis than the other, a phenomenon known as asymmetrical orientation. University of California, Berkeley biologist Ciera Martinez and colleagues examined several species of grapevines and closely related plants, which have leaves that grow in pairs, to see if both leaves in a pair tend to be oriented toward the same side (that is, have more tissue on the same side) or not. They found that opposite-side orientations are more common than same-side orientations by ratios as high as _____

10 Mark for Review

Which choice most effectively uses data from the table to complete the assertion?

(A) 646 to 1.

(B) 2.99 to 1.

(C) 3.28 to 1.

(D) 843 to 1.

P50 Value, Mean Elevation, and Rainforest Occupancy for Five New Caledonian Woody Species

Species	P50 (megapascal)	Mean elevation (m)	Rainforest occupancy (%)
Paracryphia alticola	–2.1	1,011	91
Ascarina rubricaulis	–2.28	825	86
Amborella trichopoda	–2.77	756	82
Hedycarya parvifolia	–3.19	956	86
Zygogynum crassifolium	–4.05	57	24

P50—the pressure in the xylem (water-conducting tissue) at which a plant loses 50% of its hydraulic conductivity due to embolisms (air bubbles in the water flow)—is a key index of xylem embolism resistance and, by extension, tolerance of water stress; lower P50 values indicate lower vulnerability to xylem embolisms. Studies have found that variation in this functional trait corresponds with the distribution of plant species along moisture and elevation gradients. A student hypothesizes that this pattern would persist in woody species in New Caledonia, which is home to moist tropical rainforests, dry forests, and a range of elevations. To test the hypothesis, the student analyzes P50 values as well as mean elevation and rainforest occupancy rates for New Caledonian woody species.

11 Mark for Review

Which choice best describes the extent to which the student's hypothesis is supported by data from the table?

(A) Although the data indicate that species' vulnerability to xylem embolisms varies within a relatively narrow range, their rainforest occupancy rates and elevation distributions vary much more widely; the hypothesis is therefore not supported.

(B) The data suggest a pattern in which species with enhanced tolerance of water stress are generally found at higher elevations and have higher rainforest occupancy, a relationship that holds across the majority of species examined; the hypothesis is therefore strongly supported.

(C) The data indicate that compared with *Zygogynum crassifolium*, which has the lowest rainforest occupancy rate and occurs at the lowest mean elevation, *Paracryphia alticola*, which has the highest rainforest occupancy rate and occurs at the highest mean elevation, is more vulnerable to water stress; the hypothesis is therefore strongly supported.

(D) Although the data suggest a pattern in which species' resistance to xylem embolisms tends to increase with decreasing elevation and rainforest occupancy, *Hedycarya parvifolia* is an exception to this trend;the hypothesis is therefore moderately supported.

Milena Mohri and colleagues conducted a meta-analysis of 330 studies to investigate the effect of soil characteristics, including pH, on the capacity of *Festuca rubra* and 47 other plant species to absorb chromium and other heavy metals from soil through their roots. Given that in soil, metal solubility (a key determinant of plants' metal uptake) negatively correlates with soil pH in the rhizosphere (soil immediately proximate to roots), the researchers hypothesized that plants' total metal content would negatively correlate with soil pH, but instead they found no correlation between the two variables overall. To explain the finding, they noted that many of the included studies didn't specify whether the reported soil profiles were those of the rhizosphere.

12 Mark for Review

Which statement, if true, would best account for Mohri and colleagues' explanation?

(A) Because of their distinct chemical properties, different heavy metals exhibit varying degrees of solubility response to changes in soil pH in the rhizosphere.

(B) Relative concentrations of soluble metals in the rhizosphere are more strongly mediated by soil characteristics other than soil pH.

(C) Biochemical interactions between plant roots and adjacent soil can considerably alter soil chemistry,including soil pH.

(D) Plant species vary considerably in how deeply their root systems extend beyond the topsoil layer, where soil pH and metal concentrations are typically measured.

The *Lex Orchia sumptuaria* of 182–181 BCE was one in a succession of ancient Roman sumptuary laws governing expenditures on luxury, primarily regulating large banquets and other lavish events. Such laws, enacted against a backdrop of anxiety about social dynamism, were intended to limit aspirational overreaching for status but were infrequently enforced. Moreover, enforcement was generally limited to fines, which, when levied against high-profile offenders, were widely announced. The affluent accepted fines as merely a tax on privilege, even as the amounts increased with frequent revisions to the laws. Historians have therefore argued that enforcing sumptuary laws only minimally may have been a way to avoid undermining the laws' aims, given that _____

13 Mark for Review

Which choice most logically completes the text?

(A) the few fines that were imposed under the laws were tolerable to wealthy offenders and thus did not help the laws serve their intended purpose.

(B) the public announcement of fines could signal offenders' wealth and status, placing enforcement of the laws in conflict with their intent.

(C) less-affluent citizens were reluctant to risk incurring a fine and thus voluntarily complied with the laws even in the absence of regular enforcement.

(D) authorities focused enforcement on the wealthiest offenders, recognizing that pursuing offenders who were unable to pay the fines could weaken fines' effectiveness as a deterrent.

In cave systems in the Umm Er Radhuma Formation of modern Saudi Arabia, Monika Markowska and colleagues found that from 60,000 to 530,000 years ago, both calcite and gypsum minerals formed in speleothems (stalactites, stalagmites, flowstones, and other cave formations); after this period, though, only gypsum deposits formed. The process of calcite formation requires sufficient vegetation and soil to produce carbonic acid that dissolves limestone, while gypsum deposits primarily form through evaporation processes and are typical in arid environments. Interestingly, there is some evidence of local groundwater reservoir increases after 60,000 years ago. The concurrent shift in speleothem mineral deposition, however, suggests that _____

14 Mark for Review

Which choice most logically completes the text?

(A) beginning 60,000 years ago, local precipitation fell to levels too low to maintain the vegetation necessary for gypsum formation.

(B) higher water levels in groundwater reservoirs were not due to increased local precipitation.

(C) declining local aridity after 60,000 years ago facilitated the shift in groundwater reservoir levels.

(D) changes in the mineral composition of groundwater reservoirs were independent of local precipitation patterns.

South Carolina resident Joseph Rainey, sworn in as a member of the US House of Representatives in 1870, was one of the nearly two thousand African _____ during the decade following the Civil War.

15 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) Americans' who won elections'

(B) American's who won elections

(C) Americans who won election's

(D) Americans who won elections

A single specimen of the marine protist *B. digitata* collected in the South Atlantic in 1879 served the vital role of reference organism for all _____ specific organisms as representative of an entire species for the purposes of taxonomical identification is critical, according to biologists, to resolving uncertainties that arise when classifying new specimens.

16 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) *B. digitata*, identifying

(B) *B. digitata* identifying

(C) *B. digitata* and identifying

(D) *B. digitata*. Identifying

An accomplished birder might be able to spot thousands of different species over the course of a career, from *Synthliboramphus hypoleucus* to *Fulica cristata*, but no birder has ever achieved the ultimate _____ an in-person glimpse of all 10,000-plus species of birds.

17 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) goal catching

(B) goal. Catching

(C) goal: catching

(D) goal; catching

The recognition species concept (RSC) defines species based on common fertilization systems, not genetics as in the evolutionary genetic species concept _____ instance, a population of apes that interbreed by a common fertilization system might be classified as a unique species.

18 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) (EgSC). With RSC classification, for

(B) (EgSC), with RSC classification, for

(C) (EgSC), with RSC classification. For

(D) (EgSC), with RSC classification; for

In a chemical reaction, the value known as molar mass helps convert between the mass of the reactants and the mass of the product. The gaseous _____ have molar masses of 42.08 and 150.21 g/mol, respectively.

19 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

A compounds, propylene and thymol

B compounds, propylene and thymol,

C compounds propylene and thymol

D compounds propylene and thymol,

The Alpha Centaurid meteor shower's brightness typically surpasses that of the Ursid meteor shower due to the interplay of three main factors—namely, the size and composition of the _____ velocity upon entering Earth's mesosphere, with higher-velocity meteoroids releasing more energy; and atmospheric fluctuations, such as changes in air density.

20 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) meteoroids, their

(B) meteoroids—their

(C) meteoroids; their

(D) meteoroids. Their

D-2 steel, an alloy that contains 1.50% carbon, 0.60% manganese, and 1.10% vanadium, is an air-hardening steel. _____ D-2 steel is properly quenched by heating the steel to 1,850°F and then rapidly cooling it in air to near room temperature, which increases its hardness.

21 Mark for Review

Which choice completes the text with the most logical transition?

(A) Similarly,

(B) For example,

(C) As such,

(D) Even so,

Between the thirteenth and fifteenth centuries, during the height of the Indian Ocean trade network, Ternate emerged as an important port city in the Moluccas. _____ Ternate's strategic location facilitated the exchange of cloves and nutmeg between distant regions, contributing to a prosperous commercial ecosystem that connected ports from Africa to East Asia.

22 Mark for Review

Which choice completes the text with the most logical transition?

(A) However,

(B) Indeed,

(C) Likewise,

(D) As a result,

· Since 1931, the *New York Times* (NYT) Best Sellers list has been a measure of the most successful books in the US.

· Kwame Alexander and Toni Morrison are Black authors who have reached the NYT Best Sellers list.

· *The Crossover* is a children's book by Kwame Alexander.

· It reached the NYT Best Sellers list on August 27, 2017.

· *Jazz* is a novel by Toni Morrison.

· It reached the NYT Best Sellers list on May 3, 1992.

23 Mark for Review

Which choice most effectively uses information from the given sentences to emphasize the different years in which the books reached the NYT Best Sellers list?

(A) *The Crossover* and *Jazz* are both books by Black authors that have reached the NYT Best Sellers list.

(B) Kwame Alexander and Toni Morrison are both NYT best-selling authors.

(C) Kwame Alexander's *The Crossover* and Toni Morrison's *Jazz* are both NYT best sellers, but the former is a children's book, whereas the latter is a novel.

(D) The novel *Jazz* became an NYT best seller in 1992; *The Crossover*, a children's book, joined this list as well, but in 2017.

While researching a topic, a student has taken the following notes:

· Roads designated as National Scenic Byways (NSBs) must have noteworthy features in one of six categories.
· These features can be significant for their scenic, natural, historic, cultural, recreational, or archaeological appeal.
· The Country Music Highway in Kentucky was selected to be an NSB because of the cultural features along its route.
· It provides access to theaters, music festivals, and museums.
· The Illinois River Road in Illinois was selected because of the natural features along its route.
· It provides access to river bluffs, wetlands, and prairies.

24 Mark for Review

The student wants to indicate why the two roads were selected to be NSBs. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) The Illinois River Road was chosen as an NSB for its significant natural features, while the Country Music Highway was selected for its cultural features.

(B) The Country Music Highway, like the Illinois River Road, has significant natural features such as river bluffs, wetlands, and prairies.

(C) The Country Music Highway was selected as an NSB for its noteworthy cultural features, but other NSBs could be significant for natural or archaeological reasons.

(D) The Illinois River Road provides access to noteworthy natural features but not cultural areas.

While researching a topic, a student has taken the following notes:

· 1950s: The Institute of African Studies (IAS) is proposed and initially formed at the University College of the Gold Coast (now the University of Ghana).
· Some faculty, including Peter Shinnie and his committee, support a linguistic and cultural focus for the IAS.
· This approach to African Studies is aligned with the British colonial academic models that were imposed prior to Ghana's independence in 1957.
· Other faculty resist this narrow scope, calling instead for an interdisciplinary approach that includes African history, economics, and politics.
· Early 1960s: Ghana's president Kwame Nkrumah reorganizes the IAS under an interdisciplinary, African-centered vision.
· Nkrumah's reforms are part of a broader post-1945 movement to challenge colonial systems in Africa.

The student wants to place Kwame Nkrumah's reorganization of the IAS in historical context. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Nkrumah's early 1960s reorganization of the IAS was built directly on British academic models that had long supported interdisciplinary African Studies in Ghanaian universities.

(B) Following Ghana's 1957 independence, Nkrumah reorganized the IAS with a linguistic and cultural focus, as recommended by Peter Shinnie and his committee.

(C) Nkrumah reorganized the IAS in the early 1960s to reflect an interdisciplinary, African-centered vision, aligning with a broader post-1945 movement to challenge colonial systems in Africa.

(D) Nkrumah's changes to African Studies reflected faculty concerns that the field should move beyond colonial frameworks and include areas like history, politics, and economics.

While researching a topic, a student has taken the following notes:

- A copyright prevents a book's contents from being reproduced (published) without permission from the copyright holder.
- When a book's copyright expires, the book enters the public domain and can be legally reproduced by anyone.
- *New Hampshire* is a poetry collection by Robert Frost.
- It entered the public domain in 2019.
- *When We Were Very Young* is a children's book by A.A. Milne.
- It entered the public domain in 2020.

26 Mark for Review

The student wants to emphasize a similarity between Robert Frost and A.A. Milne. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) The year was 2020, and the copyright to A.A. Milne's *When We Were Very Young* had finally expired.

(B) Both *New Hampshire* and *When We Were Very Young* are in the public domain; however, one is a poetry collection and the other a children's book.

(C) *New Hampshire*, a poetry collection by Robert Frost, entered the public domain in 2019, unlike A.A. Milne's children's book *When We Were Very Young*.

(D) Robert Frost and A.A. Milne are two authors whose works recently entered the public domain.

While researching a topic, a student has taken the following notes:

· The Clovis First hypothesis proposes that ancient humans known as Clovis were the first to populate North America.
· The Clovis arrived around 13,000 years ago.
· Stone tools found at the Saltville site in Virginia date to approximately 16,000 years ago. · Butchered mammoth bones found at the Shaffert site in Nebraska date to approximately 22,000 years ago.
· These items support the idea that humans were present in North America before the Clovis population arrived.

The student wants to emphasize a similarity between the items found at Saltville and those found at Shaffert. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Objects discovered at Saltville date to approximately 16,000 years ago; those found at Shaffert precede them, however, dating to approximately 22,000 years ago.

(B) Stone tools from Saltville date to approximately 16,000 years ago; in addition, butchered mammoth bones (which date to approximately 22,000 years ago) were discovered at Shaffert.

(C) Although the butchered mammoth bones found at Shaffert are older than the stone tools found at Saltville, both clearly predate the arrival of the Clovis population.

(D) The Clovis First hypothesis proposes that the Clovis, arriving around 13,000 years ago, were the first humans in North America.